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Name:

Teacher:.....



Pymble Ladies' College

**PRELIMINARY YEARLY EXAMINATION
2017**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using pencil for Questions 1-10.
- Write using black or blue pen for Questions 11-14. Black pen is preferred.
- Board approved calculators may be used.
- Diagrams are not drawn to scale.
- A reference sheet is provided.
- In Questions 11-14, show relevant mathematical reasoning and/or calculations.

Total Marks – 61

Section I Pages 1-4

10 marks

- Attempt all Questions 1-10
- Allow about 15 mins for this section

Section II Pages 5-9

51 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Mark	/61
Highest Mark	/61

Section I**10 Marks****Attempt all Questions**Use the multiple-choice answer sheet for Questions 1 – 10

1 Which of the following is equal to $3^x \times 2^x$?

(A) 5^x

(B) 5^{2x}

(C) 6^{2x}

(D) 6^x

2 Which of the following is the correct factorisation of $10y^2 - 19y + 6$?

(A) $(5y - 2)(2y - 3)$

(B) $(5y - 3)(2y - 2)$

(C) $(5y - 2)(3 - 2y)$

(D) $(3 - 5y)(2 - 2y)$

3 If $f(x) = \frac{x+1}{x}$, which of the following is equal to $f\left(\frac{1}{\alpha}\right)$?

(A) $1 - \alpha$

(B) $\frac{\alpha}{\alpha - 1}$

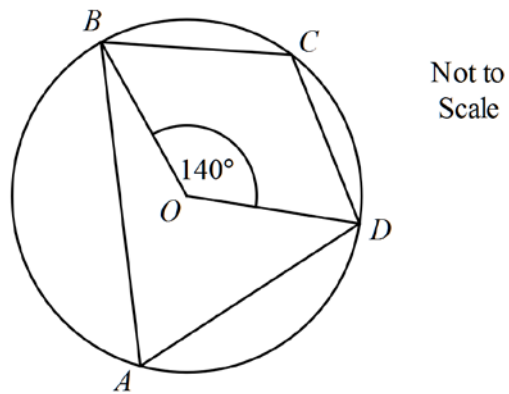
(C) $1 + \alpha$

(D) $\frac{\alpha - 1}{\alpha}$

- 4 Which of the following gives the equation of the line perpendicular to $y = 5 - 2x$ and passing through the point $(1, -3)$?

- (A) $x - 2y - 7 = 0$
- (B) $2x + y + 1 = 0$
- (C) $x - 2y + 5 = 0$
- (D) $2y + x + 5 = 0$

- 5 $ABCD$ is a cyclic quadrilateral inscribed in a circle with centre O such that $\angle BOD = 140^\circ$.

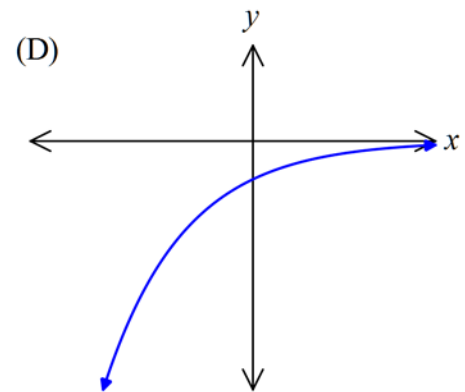
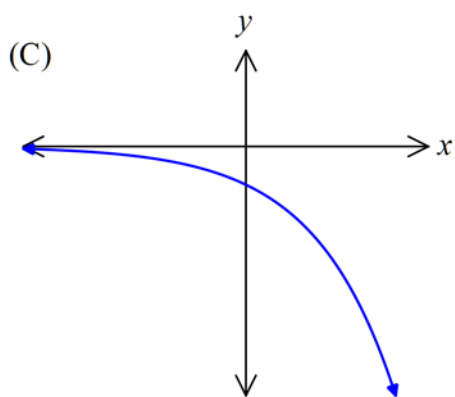
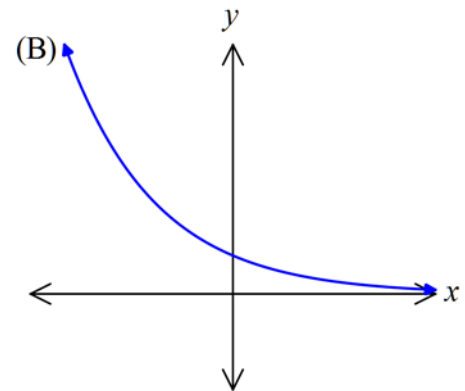
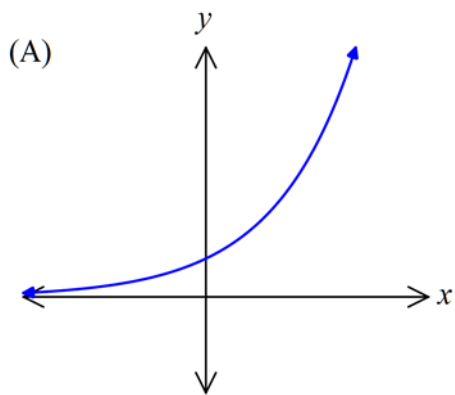


What is the size of $\angle BCD$?

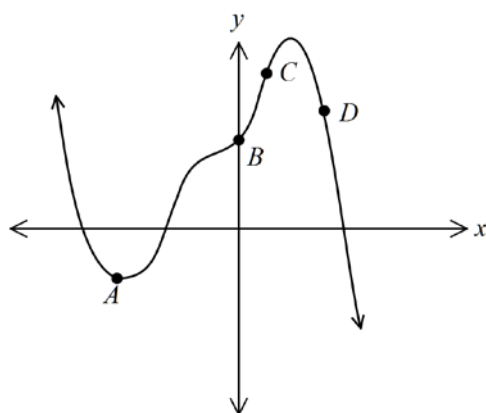
- (A) 100°
 - (B) 110°
 - (C) 120°
 - (D) 130°
- 6 What is the value of $\lim_{a \rightarrow 4} \frac{a^2 - 16}{a + 4}$?

- (A) 0
- (B) 8
- (C) -4
- (D) 4

- 7 Suppose that $f'(x) > 0$ and $f''(x) < 0$ for all real values of x . Which of the following graphs best represents $y = f(x)$?



- 8 For which point on the graph below is $f(x) > 0$, $f'(x) > 0$ and $f''(x) < 0$?

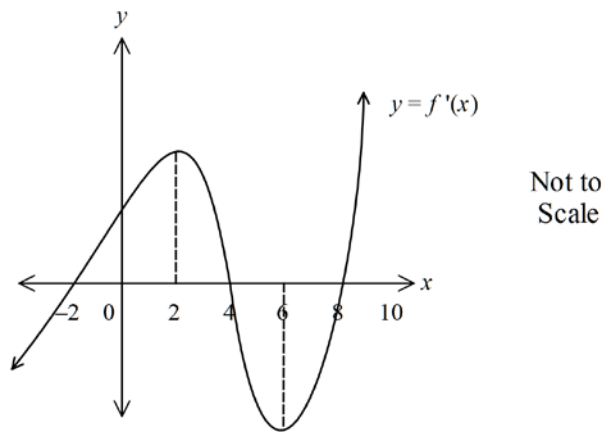


- (A) A
(B) B
(C) C
(D) D

- 9 It is known that $f''(x) = (x+4)^2(x-3)$.
How many inflexion points does the graph of $y = f(x)$ have?

- (A) 0
- (B) 1
- (C) 2
- (D) 3

- 10 The graph of the derivative $y = f'(x)$ is drawn below.



Where does a maximum turning point occur on $y = f(x)$?

- (A) $x = -2$
- (B) $x = 2$
- (C) $x = 4$
- (D) $x = 6$

Section II
51 Marks

Attempt Questions 11 – 14

Allow about 1 hour and 40 minutes for this section.

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 11 (13 marks)

- (a) Simplify $\frac{1}{p^2 - pq} - \frac{1}{pq - q^2}$. 2
- (b) Find, correct to the nearest degree, the acute angle between the lines $2x + y - 3 = 0$ and $x - 3y + 2 = 0$. 3
- (c) Find the x coordinate of the point on the curve $y = x\sqrt{x+3}$ where the tangent is parallel to the x -axis. 3
- (d) $A(8, \sqrt{200})$ and $B(1, \sqrt{72})$ are two points. Find, in simplest exact form, the coordinates of the point P which divides the interval AB in the ratio 3:1. 3
- (e) Find the domain and range of the function $f(x) = 4\sqrt{25 - x^2}$. 2

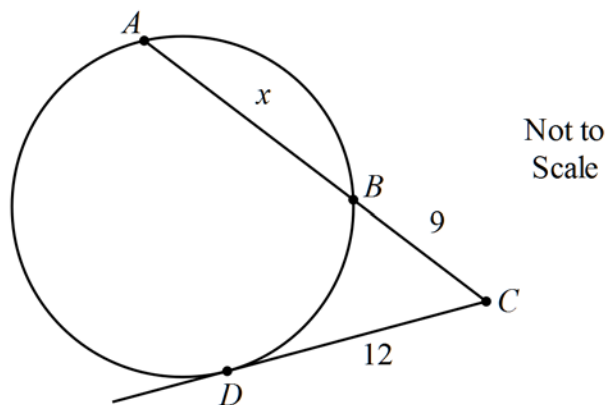
End of Question 11

Question 12 (11 marks)

(a) Solve the inequality $\frac{2x-1}{x} \geq x$. 3

- (b) Given the sequence $\sqrt{2}, \sqrt{18}, \sqrt{50}, \dots$
- (i) Show that the sequence is arithmetic. 1
- (ii) Find the value of the hundredth term. 1
- (iii) Find the sum of the first hundred terms. 1

- (c) 2



CD is a tangent at D to the circle. $AB = x$, $BC = 9$ and $CD = 12$.

Find the value of x , giving reasons.

- (d) Prove by mathematical induction 3

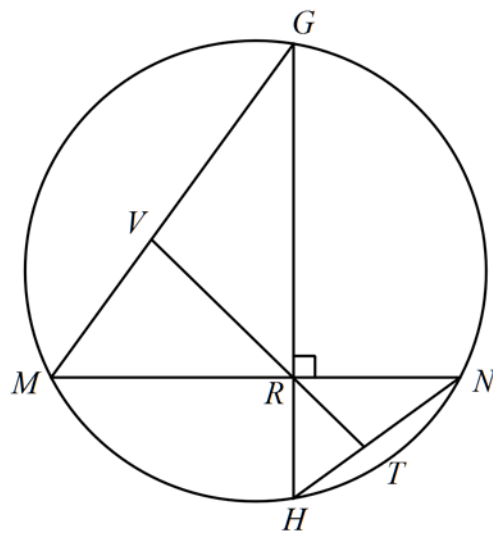
$$1 + 5 + 9 + \dots + (4n - 3) = n(2n - 1) \text{ for } n \geq 1.$$

End of Question 12

Question 13 (13 marks)

- (a) (i) Find the vertical and horizontal asymptotes of the graph $y = \frac{x-1}{3-x}$ and 4
hence, sketch the graph of $y = \frac{x-1}{3-x}$, showing intercepts with the axes.
- (ii) Hence, or otherwise, find the values of x for which $\frac{x-1}{3-x} \leq 1$. 2

(b)



Not to
Scale

MN and GH are two chords of a circle at right angles to each other, intersecting at R . From R , a line perpendicular to HN is drawn, meeting it at T . TR produced meets MG at V . Prove that:

- (i) $\triangle MVR$ is isosceles. 2
- (ii) V is the mid-point of MG . 2

Note: You do not have to copy the diagram above. It has been reproduced for you on a separate page. Insert this extra page into your answer booklet.

- (c) Prove by mathematical induction that $7^n - 1$ is divisible by 3 for all positive integers n . 3

End of Question 13

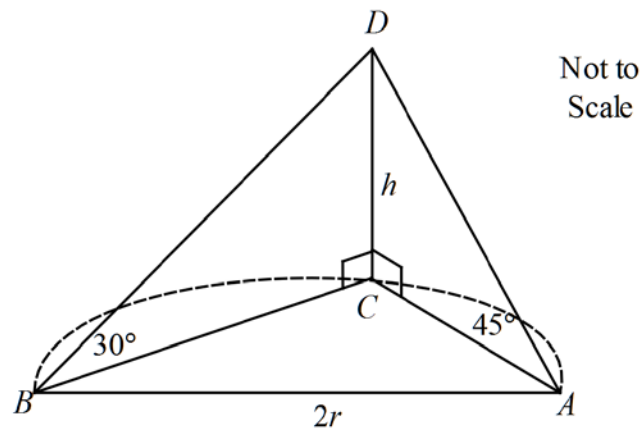
Question 14 (14 marks)

- (a) Solve $4\cos(2\alpha - 45^\circ) - 2\sqrt{3} = 0$ for the domain $0^\circ \leq \alpha \leq 360^\circ$.

3

(b)

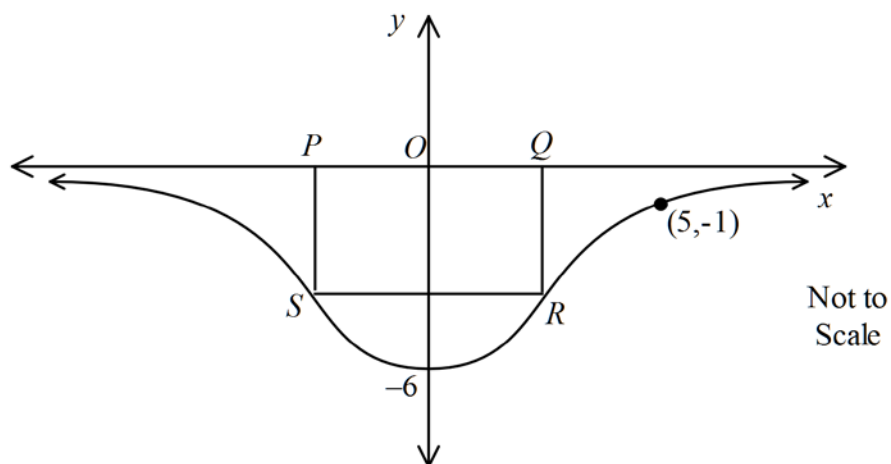
4



AB is a diameter of a semicircular piece of horizontal ground with radius r metres.
 CD is a vertical flagpole of height h metres standing with its base C on the arc AB .
From A and B the angles of elevation of the top D of the flagpole are 45° and 30° respectively. Show that $h = r$.

Question 14 continues on page 9

(c)



In the diagram above, the curve $y = \frac{h}{x^2 + k}$, where h and k are constants, has a minimum turning point at $(0, -6)$ and passes through the point $(5, -1)$. A rectangle $PQRS$ is inscribed within the curve as shown with its axis of symmetry $x = 0$.

- (i) Find the values of h and k . 2
- (ii) If Q has coordinates $(\alpha, 0)$ find the coordinates of R in terms of α . 1
- (iii) Show that the area, A , of the rectangle $PQRS$ is $A = \frac{60\alpha}{\alpha^2 + 5}$. 1
- (iv) Hence, show that the maximum area of the rectangle $PQRS$ is $6\sqrt{5}$ square units. 3

End of Paper

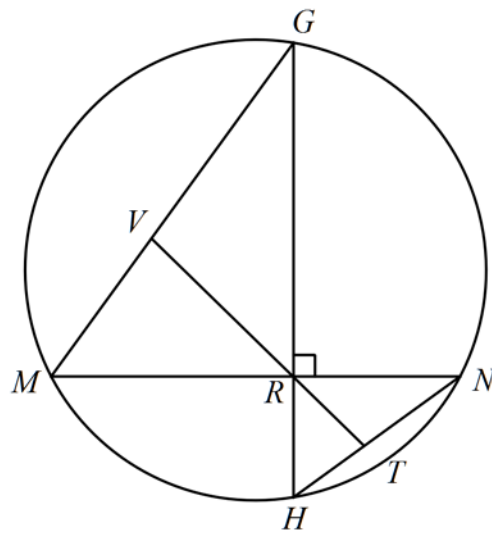
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Question 13 (b)

Note: Insert this page into your answer booklet for Question 13.



Not to
Scale

2017: 11 Ext 1 yearly exam: solutions

multiple choice

- | | |
|------|-------|
| 1. D | 6. A |
| 2. A | 7. D |
| 3. C | 8. C |
| 4. A | 9. B |
| 5. B | 10. C |

10

11 a)

$$\begin{aligned} & \frac{1}{p(p-q)} - \frac{1}{q(p-q)} \\ &= \frac{(q-p)}{pq(p-q)} \\ &= \frac{-(p-q)}{pq(p-q)} \\ &= \boxed{-\frac{1}{pq}} \end{aligned}$$

2

factorises
denom

1

correct
soln

1

b) $2x+y-3=0 \therefore m_1 = -\frac{a}{b} = -2$
 $x-3y+2=0 \therefore m_2 = -\frac{1}{-3} = \frac{1}{3}$

3

$$\begin{aligned} \therefore \tan \theta &= \left| \frac{-2 - \frac{1}{3}}{1 + (-2)(\frac{1}{3})} \right| \\ &= |-7| \end{aligned}$$

$$\begin{aligned} \therefore \tan \theta &= 7 \\ \theta &= \tan^{-1} 7 \\ &= \boxed{82^\circ} \end{aligned}$$

$$\begin{aligned} m_1 &= -2 \\ m_2 &= \frac{1}{3} \end{aligned}$$

1

correct
sub into
formula
 $\tan \theta$

1

correct
soln

1

Question 11

c) $y = x\sqrt{x+3}$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \sqrt{x+3} + \frac{x}{2\sqrt{x+3}} \\ &= \frac{2(x+3) + x}{2\sqrt{x+3}} \\ &= \frac{3x+6}{2\sqrt{x+3}}\end{aligned}$$

$$\begin{aligned}u &= x \\ u' &= 1 \\ v &= \sqrt{x+3} \\ v' &= \frac{1}{2}(x+3)^{-\frac{1}{2}} \\ v' &= \frac{1}{2\sqrt{x+3}}\end{aligned}$$

3

uses prod rule to reach

$$\frac{dy}{dx} = \sqrt{x+3} + \frac{x}{2\sqrt{x+3}}$$

simplifies to

$$\frac{dy}{dx} = \frac{3x+6}{2\sqrt{x+3}}$$

correct soln

tang || to x-axis $\therefore$ grad = 0

$$\therefore 3x+6=0$$

$$\boxed{x=-2}$$

d) $(8, \sqrt{200})$ $(1, \sqrt{72})$

3:1

3

$$x = \frac{8(1)+3(1)}{4}$$

$$y = \frac{\sqrt{200}(1)+3\sqrt{72}}{4}$$

correct simplified soln

$$x = \frac{8(1)+3(1)}{4}$$

$$= \frac{11}{4}$$

$$y = \frac{\sqrt{200}(1)+3\sqrt{72}}{4}$$

$$y = \frac{10\sqrt{2}+18\sqrt{2}}{4}$$

$$y = \frac{28\sqrt{2}}{4}$$

$$\therefore P \text{ is } \left(\frac{11}{4}, 7\sqrt{2}\right)$$

e) $f(x) = 4\sqrt{25-x^2}$

$$D: 25-x^2 \geq 0 \quad -5 \quad 5$$

$$(5+x)(5-x) \geq 0$$

$$\therefore \boxed{-5 \leq x \leq 5}$$

$$R: \boxed{0 \leq y \leq 20}$$

2

$$D: -5 \leq x \leq 5$$

$$R: 0 \leq y \leq 20$$

Question 12

a) $\frac{2x-1}{x} \geq x, x \neq 0$

$\times x^2$ $x(2x-1) \geq x^3$

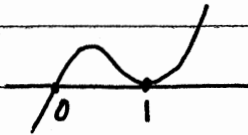
$$x(2x-1) - x^3 \geq 0$$

$$x[2x-1-x^2] \geq 0$$

$$x(-x^2+2x-1) \geq 0$$

$\div -1$ $x(x^2-2x+1) \leq 0$

$$x(x-1)^2 \leq 0$$



$$x \leq 0, x = 1 \text{ but } x \neq 0$$

$\therefore$ Sol'n: $x < 0, x = 1$

mult by x^2

reaches
 $x(x-1)^2 \leq 0$
or equivalent

correct
sol'n

* Alternate sol'n
using critical points

- ① x by x .
② Eq'n to find 2 sol'n
③ Test ③ correct sol.

b) series = $\sqrt{2} + 3\sqrt{2} + 5\sqrt{2} + \dots$

i) $T_2 - T_1 = 3\sqrt{2} - \sqrt{2} = 2\sqrt{2}$

$T_3 - T_2 = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$

$\therefore$ AP with $a = \sqrt{2}, d = 2\sqrt{2}$

* Wrong if
straight to decimal

$\rightarrow$ Leave in
exact form.

R/W

ii) $T_n = a + (n-1)d$

$T_{100} = \sqrt{2} + 99(2\sqrt{2})$

$= 199\sqrt{2}$

R/W

iii) $S_n = \frac{n}{2}[a + l]$

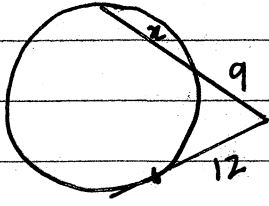
$= \frac{100}{2}[\sqrt{2} + 199\sqrt{2}]$

$= 50(200\sqrt{2})$

$= 10000\sqrt{2}$

R/W

c)



$9(9+x) = 12^2$
 (product of intercepts
 of secant equals
 tangent squared)

$$81 + 9x = 144$$

$$x = 7$$

2

$$9(9-x) = 144$$

reason

*Reason must
 include underlined
 words or equivalent

d) let $S(n)$ be the statement
 $1+5+9+\dots+(4n-3) = n(2n-1)$, $n \geq 1$

STEP 1: For $n=1$, LHS = 1 RHS = $1(2(1)-1)$
 $= 1$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore S(1) \text{ is true}$$

STEP 2: Assume true for $S(k)$
 i.e. $1+5+\dots+(4k-3) = k(2k-1)$, $k \geq 1$

STEP 3: Prove true for $S(k+1)$

i.e. RTP:

$$1+5+\dots+(4k-3)+(4k+1) = (k+1)(2k+1)$$

$$\text{LHS} = 1+5+\dots+(4k-3) + (4k+1)$$

$$= k(2k-1) + (4k+1) \text{ by induction hypothesis [I]}$$

$$= 2k^2 - k + 4k + 1$$

$$= 2k^2 + 3k + 1$$

$$= (k+1)(2k+1)$$

$$= \text{RHS}$$

STEP 4: $\therefore$ if $S(k)$ is true, then $S(k+1)$ is true

Since $S(1)$ is true, it follows by
 mathematical induction that
 $S(n)$ is true for $n \geq 1$. QED.

$$S(1) \text{ true}$$

$$S(k) \text{ and}$$

$$S(k+1)$$

$$\text{statements}$$

$$\text{correct}$$

$$\text{correct}$$

$$\text{proof}$$

$$\text{[I] logical}$$

$$\text{mi}$$

$$\text{statement}$$

$$(\text{be v.}$$

$$\text{generous})$$

Question 13

a) i) $y = \frac{x-1}{3-x}$

zeros: $x=1$

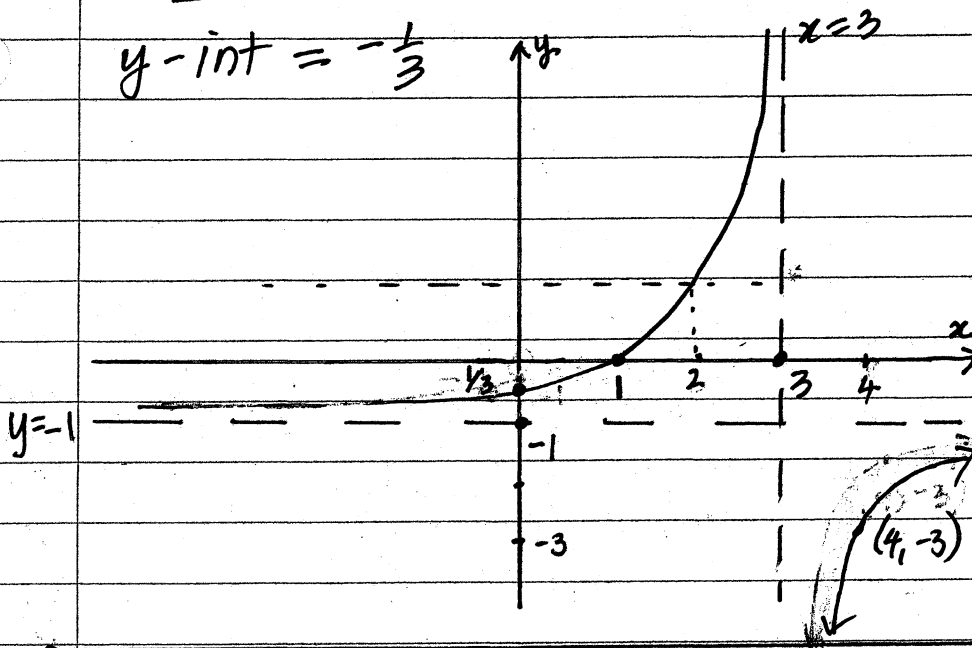
discontinuities: $x=3$

$\therefore$ vertical asymptote at $x=3$

$\div x \therefore y = \frac{1 - \frac{1}{x}}{\frac{3}{x} - 1}$ as $x \rightarrow \infty, y \rightarrow -1$

$\therefore y = -1$ is horizontal asymp.

$y\text{-int} = -\frac{1}{3}$



vert. asymptote
 $x=3$

$y=-1$ horiz
ass with
correct limit
notation

$y\text{-int}$
 $x\text{-int}$
 $x=3$
 $y=-1$

correct
shape &
correct
quads
+ point on
lower arm.

ii) soln: $\frac{x-1}{3-x} = 1$

$x-1 = 3-x$

$2x = 4$

$x = 2$ $\therefore$ from graph

$\therefore$ soln: $x \leq 2, x > 3$

2

$x \leq 2$ | 1

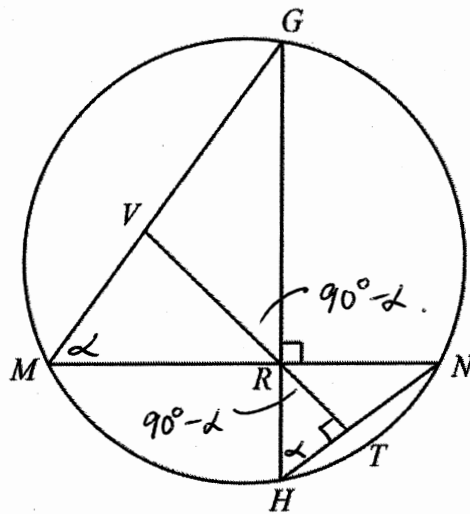
$x > 3$ | 1

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Question 13 (b)

Note: Insert this page into your answer booklet for Question 13.

Not to
Scale(1) Let $\angle GMN = \alpha$
 $\angle GHN = \angle GMN$ (angles at circumference standing on same arc GN) 1
 $\angle RHT = \angle GHN = \alpha$ (common) $\angle VMR = \angle GMN = \alpha$ (common) $\angle HTR = 90^\circ$ ($RT \perp HN$) $\angle HRT + \alpha + 90^\circ = 180^\circ$ (angle sum of $\triangle HRT$) $\angle HRT = 90^\circ - \alpha$
 $\angle GRV = \angle HRT$ (vertically opposite)
 $= 90^\circ - \alpha$
 $\angle VRM + (90^\circ - \alpha) = 90^\circ$ (adjacent complementary angles where $\angle MRG = 90^\circ$)
 $\angle VRM = \alpha$
 $\therefore \angle VMR = \angle VRM = \alpha$ $\therefore \triangle MVR$ is isosceles (two equal angles)

13b)(ii) In $\triangle VRG$

$$\begin{cases} \angle GRV = 90^\circ - \alpha & (\text{from (i)}) \\ \angle GVR = \alpha + \alpha & (\text{exterior angle of } \triangle MVR) \\ & = 2\alpha \\ \textcircled{1} \quad \angle VGR + 2\alpha + (90^\circ - \alpha) = 180^\circ & (\text{angle sum of } \triangle VRG) \\ & \therefore \angle VGR = 90^\circ - \alpha \\ & \therefore \angle VGR = \angle GRV = 90^\circ - \alpha \\ & \therefore VR = VG & (\text{sides opposite equal angles}) \end{cases}$$

$$\begin{cases} \text{In } \triangle MVR \\ \textcircled{1} \quad \begin{cases} VR = VM & (\text{sides opposite equal angles}) \\ \therefore VM = VG \\ \therefore V \text{ is midpoint of } MG \end{cases} \end{cases}$$

Question 13

c) Let the statement be

 $7^n - 1$ is divisible by 3
for integer $n \geq 1$
Prove statement true for $n=1$

$$7^1 - 1 = 6$$

$$= 3 \times 2 \quad \text{which is divisible by 3}$$

 $\therefore$ true for $n=1$ statement
true for $n=1$

1

correctly
states k
hypothesis
and $k+1$
statement

2

correct
solution

3

Assume statement true for $n=k$, for integer $k \geq 1$ ie assume $7^k - 1 = 3P$ for integer P

$$\text{ie } 7^k = 3P + 1$$

Prove statement true for $n=k+1$ ie prove $7^{k+1} - 1 = 3Q$ for integer Q

$$7^{k+1} - 1 = 7 \times 7^k - 1$$

$$= 7 \times (3P + 1) - 1 \quad (\text{by assumption})$$

$$= 21P + 7 - 1$$

$$= 21P + 6$$

$$= 3(7P + 2)$$

$$= 3Q$$

where $Q = 7P + 2$ is an integer if P is an integerIf statement is true for $n=k$, then it is also true for $n=k+1$

Since statement is true for $n=1$, then it is also true for $n=1+1=2$, $n=2+1=3$, and so on for all integers $n \geq 1$

Question 14

a) $4 \cos(2a - 45) = 2\sqrt{3}$

$$\cos(2a - 45) = \frac{\sqrt{3}}{2}$$

let $u = 2a - 45^\circ$

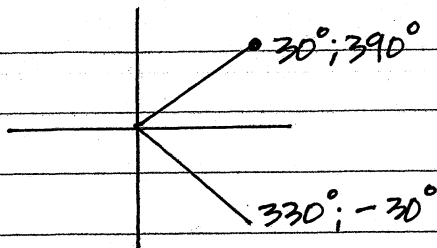
$$\cos u = \frac{\sqrt{3}}{2}$$

related $u = 30^\circ \checkmark$

$$0^\circ \leq a \leq 360^\circ$$

$$-45^\circ \leq 2a - 45 \leq 675 \quad (3)$$

$$\therefore -45^\circ \leq u \leq 675^\circ \checkmark$$



$$\therefore 2a - 45 = -30^\circ; 30^\circ; 330^\circ; 390^\circ$$

$$2a = 15^\circ; 75^\circ; 375^\circ; 435^\circ$$

$$\therefore a = 7.5^\circ; 37.5^\circ; 187.5^\circ; 217.5^\circ \checkmark$$

recognises domain

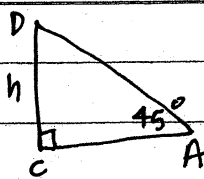
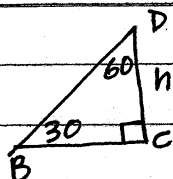
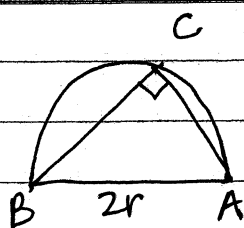
$$-45^\circ \leq u \leq 675^\circ \quad |$$

related angle
 30° 2

correct
soln 1

• poorly done.
• few did
necessary domain
amendment

b)



in $\triangle BDC$: $\frac{BC}{h} = \tan 60$

$$BC = h \tan 60$$

$$BC = \sqrt{3}h \quad \text{--- [1] } \checkmark$$

in $\triangle DCA$: $DC = AC$ (sides opposite equal angles: isosceles $\triangle$)

$$\therefore AC = h \quad \text{--- [2]}$$

in $\triangle ABC$: $\angle BCA = 90^\circ$ (angle in semicircle)

$$\therefore (2r)^2 = (\sqrt{3}h)^2 + (h^2) \quad \checkmark$$

$$4r^2 = 3h^2 + h^2$$

$$r^2 = h^2$$

$$\therefore \boxed{r = h} \quad \checkmark \text{ since } h > 0, r > 0$$

(4)

$$BC = \sqrt{3}h$$

or

$$AC = h$$

$$\angle ACB = 90^\circ$$

+ reason 1

realises eqn

$$BA^2 = BC^2 + AC^2$$

or

uses cos

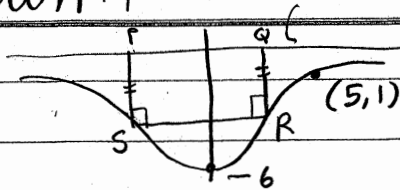
rule

correctly 1

corr soln 1

Question 14

(c)



i) $y = \frac{h}{x^2+k}$ sub (0, -6)

$$\therefore -6 = \frac{h}{k} \quad \text{--- [1]} \quad \therefore h = -6k$$

sub (5, -1) $\therefore -1 = \frac{h}{25+k}$
 $\therefore h = -25-k$ --- [2]

[1] $\rightarrow$ [2] $-6k = -25-k$
 $-5k = -25$

$$k = 5$$

$$\therefore h = -6(5)$$

$$h = -30$$

2

substs either (0, -6) or (5, -1)	1mk
correct soln	2

(ii)

$$y = \frac{-30}{x^2+5} \quad Q(a, 0) \quad R(a, y)$$

$$y = \frac{-30}{a^2+5}$$

$$R\left(a, \frac{-30}{a^2+5}\right)$$

1

subst $x=a$ into eqn	1
----------------------	---

iii) $PQ = 2a$, $QR = \frac{30}{a^2+5}$
 $\therefore \text{area} = 2a \cdot \left(\frac{30}{a^2+5}\right)$

$$\text{area} = \frac{60a}{a^2+5} \quad \text{as req'd}$$

1

$A = 2a \left(\frac{30}{a^2+5}\right)$	1mk
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Question 14

11

iv)

$$A = \frac{60d}{d^2+5}$$

candidates are encouraged to state these conditions

For max area, $\frac{dA}{dd} = 0$, $\frac{d^2A}{dd^2} < 0$

$$u = 60d$$

$$u' = 60$$

$$v = d^2+5$$

$$v' = 2d$$

say why we want

$\frac{dA}{dd} = 0$ to zero.

$$\therefore \frac{dA}{dd} = \frac{60(d^2+5) - 120d^2}{(d^2+5)^2}$$

$$\frac{dA}{dd} = \frac{300 - 60d^2}{(d^2+5)^2}$$

for $\frac{dA}{dd} = 0 \therefore 60d^2 = 300$

$$d^2 = 5$$

$$d = \sqrt{5}, d \geq 0$$

sign of $\frac{dA}{dd}$

d	2	$\sqrt{5}$	3
$\frac{dA}{dd}$	$\frac{20}{27}$	0	$-\frac{60}{49}$

used $\frac{d^2A}{dd^2}$

$$\frac{d^2A}{dd^2} = \frac{-120d(d^2+5)^2 - 4d(d^2+5)(300-60d^2)}{(d^2+5)^4}$$

when $d = \sqrt{5}$, $\frac{d^2A}{dd^2} = -2.6832815... < 0$

since grad changes from pos to neg $\therefore d = \sqrt{5}$ gives max area

$\therefore d = \sqrt{5}$ gives max area

$$\therefore A = \frac{60\sqrt{5}}{(\sqrt{5})^2+5} = \frac{60\sqrt{5}}{10}$$

$$A = 6\sqrt{5} \text{ u.s. as required}$$

3

gets $\frac{dA}{dd} = \frac{300-60d^2}{(d^2+5)^2}$ 1m

solves $\frac{dA}{dd} = 0 \therefore d = \sqrt{5}$ 2m

shows max and subst. into area to get $A = 6\sqrt{5}$ 3

must explain how $d = \sqrt{5}$ gives a max area

sign line $\frac{d^2A}{dd^2}$